

Invited paper

Celebrating 40th SEDOPTICA anniversary

Recent advancements in stimulated-Brillouin-scattering slow light

Avances recientes en luz lenta por scattering Brillouin estimulado

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ABSTRACT:

A review on the work performed by our group on slow light via Stimulating Brillouin scattering (SBS) is presented. We describe the fundamentals of the slow-light phenomena and how SBS can be used to obtain tunable optical delays in optical fibers. We then review the mechanisms to overcome the limitations imposed by the narrow SBS resonance bandwidth and to optimize the slow-light performance for data rates used in modern telecommunication networks. We also describe the process of coherent light storage in optical fibers based on SBS which paves the way to optical buffering applications. Finally, future perspectives of this exciting field of research are discussed.

Key words: Slow Light, SBS, Fiber optics.

RESUMEN:

En la presente revisión se exponen los resultados obtenidos por nuestro grupo en luz lenta vía scattering Brillouin estimulado (SBS por sus siglas en inglés). En ella describimos los fundamentos de los fenómenos de luz lenta y cómo el proceso de scattering Brillouin estimulado puede ser utilizado para obtener retardos sintonizables en fibras ópticas. Posteriormente expondremos los mecanismos utilizados para superar las limitaciones impuestas por el estrecho ancho de banda de la resonancia Brillouin y para optimizar los resultados a tasas de datos correspondientes a las redes de telecomunicaciones actuales. A continuación también describimos el proceso de almacenamiento coherente de la luz en fibras ópticas basado en SBS, con aplicaciones en buffers completamente ópticos. Finalmente, se comentan las perspectivas futuras de este excitante campo de investigación.

Key words: Luz lenta, SBS, Fibras ópticas.

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1. Introduction

Slow light has become a very promising field of research to develop technologies that can replace the electronic stages in telecommunications systems. The development of tunable optical delay lines may have applications in many branches of optical technology. The construction of optical buffers, optical switchers or optical synchronizers

are some devices that could be implemented in present telecommunications networks [1,2]. Other applications, such as increasing the sensitivity of spectroscopic interferometers [3,4], or enhancement of nonlinear optical phenomena [5, 6] has also been demonstrated.

"Slow (Fast) light" refers to a broad class of phenomena where the propagation of a light pulse

inside a medium is altered to produce abnormally slow (fast) group velocities. This is achieved by exploiting the change of optical dispersion near narrow spectral resonances. Due to the Kramers-Kronig relations [7], a sharp spectral change in the optical transmission spectrum of a medium results into a steep quasi-linear variation of the index of refraction. Recalling the definition of the group velocity,

$$v_g = \frac{c}{n + \omega \frac{dn}{d\omega}} = \frac{c}{n_g}, \quad (1)$$

where n_g is the group index, we see that a rapid spectral variation of the index of refraction leads to a very large value of n_g when $dn/d\omega > 0$, i.e. normal dispersion. When $n_g \gg 1$, a dramatic reduction of the group velocity can be achieved. Velocities as small as several m/s have been reported [8]. Equation (1) also predicts that, for the case of anomalous dispersion ($dn/d\omega < 0$), the group velocity can take values larger than c or even negative, which is known as *Fast Light*.

Although ideas on slow and fast Light can be found more than 100 years ago, the work that triggered the recent research was the demonstration by Hau *et al.* in 1999 of group velocities as slow as 17 m/s in Bose-Einstein condensates [8]. To produce this result, electromagnetically induced transparency (EIT) was used. In this phenomenon, a control beam illuminates a dilute gas of atoms creating a narrow transparency window in the absorption of another atomic transition that shares a common energy level. Although this work showed an impressively small value of v_g , the limitation of the ultracold temperatures needed to achieve it pushed the research to find other ways to produce slow light at room temperature for practical applications.

Soon thereafter, different optical processes have been shown to produce slow light at room temperature. Coherent population oscillations (CPO) [9–11], stimulated scattering [12–14], optical parametric amplification [15,16], conversion-dispersion [17,18], or fiber Bragg gratings [19,20] are a few techniques. Among them, optical fibers provide a very attractive scenario for practical applications due to their easy integration with existing communications networks. Besides that, the delay achievable scales with the propagation distance in the medium, which can be as long as several kilometers in optical fibers due to their low-loss characteristics.

This review focus on the research on slow light in optical fibers performed by our group and our

collaborators. In particular, we describe how slow light can be achieved by Stimulated Brillouin Scattering (SBS) in transparent optical fibers and how it can be optimized to produce large delays at high data rates. In Section 2, we describe the basics of SBS and how it can be used to obtain slow light. Section 3 is devoted to the mechanisms used to extend the performance to larger bandwidths. Section 4 describes an optimization method that minimizes the pulse distortion while maximizing the delay. Section 5 explains the process of coherent optical storage via SBS. Finally, Section 6 is devoted to summarize the work on SBS slow light and describes future perspectives.

2. Slow light by SBS

Stimulated Brillouin scattering has proven to be an effective way to produce slow light in optical fibers at room temperature. Regarding its advantages over other technologies, it is relatively simple to implement, can be obtained at any wavelength within the transparency window of the optical fiber, only needs moderate pump powers, and can use off the shelf fiber optics components providing easy integration with existing telecommunications systems. These advantages have made SBS a good candidate for practical applications.

Stimulated Brillouin scattering in optical fibers takes place when a pump field with frequency ω_p and a counterpropagating probe field with frequency ω_s interact through their mutual interaction with a longitudinal acoustic wave with frequency Ω . The value of the acoustic frequency depends on the properties of the material, and is ~ 10 GHz for standard single-mode-silica fibers. The acoustic wave causes a modulation of the index of refraction, which results in a scattering of power into or out of the probe beam. The beating between both fields enhances the acoustic wave by a process called electrostriction, or the tendency of the material to compress in the presence of a strong electric field, thereby reinforcing the scattering. This coupling is efficient only if both the energy and momentum are conserved, which is satisfied when the two optical waves are counterpropagating and have a frequency difference that is resonant with the acoustic wave, i.e., $\omega_s = \omega_p \pm \Omega$. If the frequency of the probe beam is downshifted (upshifted) from the pump field frequency, lying in the so-called Stokes (antiStokes) sideband, the probe beam experiences exponential amplification (absorption). The gain and absorption features are narrowband and Lorentzian-shaped, with a full-width at half maximum (FWHM) of ~ 40 MHz. Figure 1 shows the gain, refractive index and group

index for the amplification resonance ($\omega_s = \omega_p - \Omega$). As can be seen in Fig. 1(a), large normal dispersion takes place at the center of the resonance, which leads to an increase of the group index (see Fig. 1(b)) and therefore, to a reduction of the group velocity.

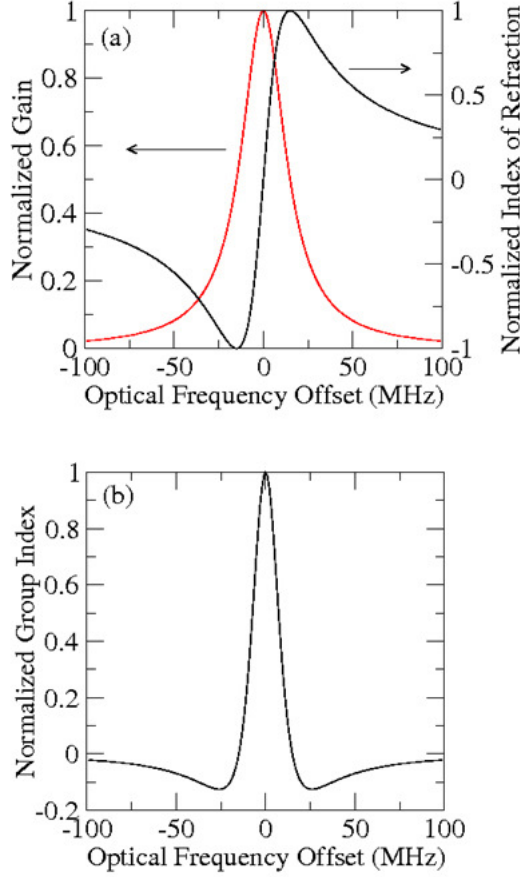


Fig. 1: (a) Normalized gain and index of refraction of the SBS resonance as a function of the optical frequency. (b) Normalized group index of the resonance.

The time delay experienced by a pulse propagating along the optical fiber is directly proportional to the group index change. For a signal whose carrier frequency is placed at the center of the resonance, the delay can be expressed as

$$\Delta T = \Delta n_g \frac{L}{c} = \frac{G}{\Gamma_B}, \quad (2)$$

where $G = g_0 I_p L$ is the small-signal gain at the center of the resonance and $\Gamma_B/2\pi$ is the FWHM linewidth of the Brillouin gain spectrum. The small-signal gain is proportional to the Brillouin gain parameter g_0 , to the propagation length L , and to the pump intensity I_p , which allows to continuously control the time delay. Slow light via SBS was first proposed by Gauthier in 2004 [22], and demonstrated independently by Song *et al* [13] and

by Okawachi *et al.* [12] in 2005. Figure 2 shows two examples of delayed pulses obtained in Okawachi's experiment. In Fig. 2(a) a 63-ns-long pulse is delayed 25 ns, which corresponds to a fractional delay (defined as the ratio of the time delay and the temporal pulse width) of 0.4. This delay was accompanied by a small amount of broadening, with an output-to-input pulse width-ratio of 1.05. Larger relative delay could be achieved when a shorter pulse was delayed. Figure 2(b) shows a 15-ns-long pulse delayed by 20 ns, obtaining a fractional delay of 1.3. However, this increase of the relative delay comes at the price of larger broadening. In this case, an output-to-input pulse width-ratio of 1.4 was measured.

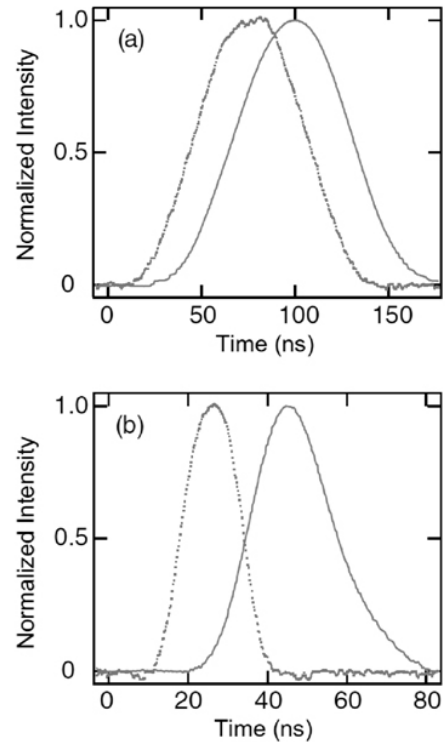


Fig. 2: Input signal pulse (dotted) and delayed pulse (solid) for a gain parameter $G=11$ and for two input pulse widths: (a) 63 ns long and (b) 15 ns. Reprinted figure with permission from Ref. [12]. Copyright 2005 by the American Physical Society.

Although SBS slow light has many advantages, it also has some limitations. First, there is an upper limit in the value of the gain parameter G . If $G \approx 15$, spontaneous Brillouin scattering begins to deplete the pump and saturate the probe gain and, therefore, the time delay. Besides that, the small linewidth of the Brillouin gain induces strong distortion when the pulse bandwidth is larger than ~ 40 MHz. This limits the data rates that can be propagated without severe signal degradation to a few Mb/s. The first limitation can be avoided using multiple delay-lines and inserting optical

attenuators between them to prevent gain saturation, as it was shown by Song *et al* [23]. In order to overcome the second limitation, a larger Brillouin linewidth must be used. The next section describes how can this be done to obtain slow light in fibers at high-data rates using SBS.

3. Broadband SBS slow light

The small Brillouin linewidth for a monochromatic pump beam, which is ~40 MHz in standard optical fibers, limits the pulse bandwidth that can be delayed by the slow light system. As can be seen in Fig. 2, pulse distortion increases as the duration of the pulse is reduced, degrading the transmitted signal. This imposes a data-rate limit of ~40Mb/s. In order to develop applications for modern communication systems, bitrates of 10 Gb/s with a tolerable distortion are required.

In order to reduce the pulse distortion, reshaping of the SBS gain profile must be carried out. This can be fulfilled using a polychromatic pump beam. In this case, each of the monochromatic waves in which the pump spectrum can be decomposed generates a Lorentzian Brillouin gain, which delays the corresponding part of the probe spectrum that lies within it. The resultant gain profile is then the convolution of the pump spectrum with the Lorentzian Brillouin gain,

$$g(\nu) = g_B(\nu) \otimes I_p(\nu), \quad (3)$$

which closely matches the pump spectrum profile when its width is much larger than the natural Brillouin resonance width. Stenner *et al.* [24] were the first to show that distortion can be reduced using two closely spaced monochromatic pump beams. The gain profile generated by this dual frequency pump spectrum flattens the gain over a finite bandwidth, thereby reducing the filtering effect caused by a frequency-dependent gain. Since their work, different studies have followed this approach to minimize pulse distortion. Schneider *et al* [25] showed a ~30% reduction of the pulse broadening using three different Brillouin gain lines in a two-stage-delay-line. Shi *et al* [26] showed an increase by a factor of two in the fractional delay at three times larger modulation bandwidth for an optimized triple-gain-line medium as compared with a simple-gain-line. More recently Sakamoto *et al* [27] used 20 discrete line pump spectra to suppress the broadening factor (defined as the output-to-input pulse-width ratio) to 1.19 for 5.44-ns-long pulses in a two-stage-delay-line. Nevertheless, all of these works rely on the use of an external phase or intensity modulator to generate

multiple pump spectral lines. This limits the maximum bandwidth achievable to hundreds of MHz with present-day commercially-available modulators. The maximum value obtained up to now using this approach is ~330 MHz [28].

However, the SBS gain spectrum can be continuously broadened using a modulated pump beam. Herráez *et al.* [29] were the first to increase the SBS gain bandwidth up to ~325 MHz by modulating the pump injection current with a Gaussian noise source. The bit-rate of the modulating function was set fast enough to ensure that the signal pulse experiences an homogeneous pump spectrum when integrated over the whole fiber length. Zhu *et al* [30] extended their results to achieve a bandwidth of 12.6 GHz, thereby supporting data rates over 10 Gb/s. In this study, 47 ps of delay for 75-ps-long pulses was obtained with a pulse broadening factor of 1.4. Figure 3 shows the measured SBS gain and absorption spectrum for Zhu's experiment, and the delay and distortion performance as a function of the maximum gain. The absorption spectrum nearly overlaps with the gain profile, which causes further broadening of the SBS gain to reduce the slow-light efficiency. This limitation was overcome by Song *et al* [31] using a second broadened pump separated from the first one by the acoustic wave frequency. In this situation, the gain resonance generated by this second pump compensates the absorption caused by the first pump, allowing to use of a wider pump spectra to increase the SBS gain bandwidth. In their work, a 25 GHz gain-bandwidth was obtained using this approach.

Using a broadened pump spectrum to support multigigabit per second data rates bring applications a little bit closer. Nevertheless, increasing the SBS gain bandwidth to accommodate shorter pulses comes at the cost of requiring a higher pump power to achieve the same gain (and the same delay). This results in very high pump powers in order to obtain relative delays larger than one while keeping a small pulse broadening. For example, pump powers of ~600 mW were necessary to achieve a fractional delay of 0.6 for 75-ps-long pulses with a pulse broadening factor of 1.4 in Zhu's experiment. Besides that, the frequency-dependent gain of a broadened Gaussian-shaped SBS gain still introduces some distortion in the delayed pulses, as can be seen in Fig. 3. Therefore, a natural question arises. Is there any way to reduce the distortion and increase the delay at the same time? In the next section, we describe a method to optimize the SBS gain spectrum in order to increase the delay and reduce the pulse broadening.

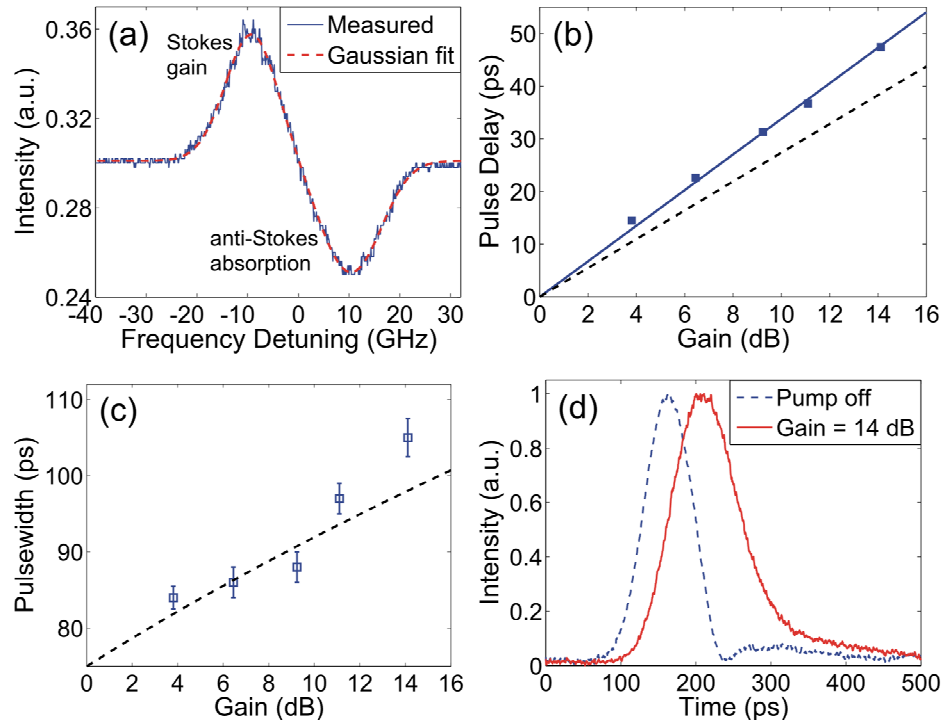


Fig. 3. (a) Measured SBS gain and absorption spectrum with a dual Gaussian fit. The SBS gain bandwidth is 12.6 GHz (FWHM). Pulse delay (b) and pulse width (c) as a function of the SBS maximum gain. In (b) the solid line is the linear fit of the experimental data (solid squares), and the dashed line is the theoretical prediction without considering the anti-Stokes absorption. In (c), the dashed curve is the theoretical prediction. (d) Pulse waveforms at 0 dB and 14 dB SBS gain. The input pulsewidth is 75 ps. Reprinted figure with permission from Ref. [30]. Copyright 2006 IEEE.

4. Optimized SBS gain profile

Tailoring the SBS gain spectrum by using a polychromatic pump beam gives SBS slow light greater flexibility. As shown in the previous section, when the pump linewidth is much larger than the natural Brillouin resonance width, the resultant gain profile closely matches the pump spectrum. This feature opens the possibility to shape the spectral resonance in order to optimize the slow light performance. An optimized SBS gain profile must maximize the delay while minimizing the distortion experienced by the delayed pulses. As mentioned above, Stenner *et al* [24] showed that the main cause for distortion is the frequency-dependent gain, whereby each frequency component of the signal pulse experiences different amplification, which leads to a distorted pulse upon propagation through the fiber. Therefore, a flat-top SBS gain profile is needed to minimize the distortion. Maximizing the delay requires an increase in the variation of the index of refraction in the resonance, maximizing the value of the group index. Zadok *et al* [32] demonstrated that a larger change in the index of refraction can be accomplished by designing a pump spectrum with sharp edges. They showed an increase of the delay by a factor of 30% compared with a Gaussian pump

profile. In order to design this pump spectrum, they produced a specific form of the modulating function of the injection current of the pump laser, which allowed them to control the way in which the pump spectrum is broadened. Yi *et al* [33] also showed that a super-Gaussian gain profile reduced the filtering effect because the edges of the gain profile are sharpened. In this case, a super-Gaussian noise distribution was used to create a pump spectrum profile with sharp edges. Their experiments focused on the delay performance of a SBS slow light system when a pseudo-random bit sequence of pulses are injected into the system. They showed that propagation of a bit stream without errors introduced by the delay line could be achieved for 10 Gb/s random sequences, obtaining a maximum time delay of 35 ps under this constraint. This result can be improved by using other modulating format with higher dispersion tolerance and spectral efficiency, such as phase shifted binary transmission (PSBT) or differential phase shift keying (DPSK) [33, 34]. Pant *et al* [35] recently addressed theoretically the question of finding the optimal broadband SBS gain profile under the constraint of maximum allowed power. They found that the optimal gain profile is essentially a flat-top spectrum with sharp edges,

surrounded by narrow absorption features. They also found that these features are no longer needed for bandwidths above 7 GHz. More recently, Schneider [36] obtained basically the same result, showing an enhancement of the time delay when two narrow losses are placed at the wings of the gain. While the flat-top profile minimizes the distortion, the sharp edges maximizes the group index, and hence, the delay.

Experimental demonstration of this optimum broadband SBS gain profile has been recently carried out by our group [37]. In this work, optimization of the modulating function to tailor the pump spectrum was performed following the approach of Zadok *et al* [32]. Initially, a modulating waveform is determined by measuring the transient behavior of the instantaneous frequency of the pump laser, following the work of Shalom *et al* [38]. Then, adjustments to the waveform are made using an iterative scheme to compensate the deviations of the gain spectrum from the ideal profile. Figure 4 shows the measured gain profile with a FWHM of 9.2 GHz and the delay performance for 100-ps-long pulses (which correspond to 10 Gb/s data rates), expressed as fractional delay, already defined previously.

A comparison with the theoretical predictions for a Gaussian profile of the same bandwidth can also

be seen. The optimized gain profile improves the time delay by a factor of 1.3 in comparison to the expected results for the Gaussian profile, while the distortion is reduced by a factor of 1.2.

5. Stored light via SBS

A strong motivation of slow-light research is to obtain an all-optical memory where the storage time can be continuously adjustable. As the data remains in the optical domain an all-optical buffer can increase the efficiency of routers in telecommunications networks, avoiding the high power consumption and the bandwidth limitations of electronic buffers. One of the first attempts to fulfill this goal in the context of slow light was the so-called stored or stopped light in EIT. Here, the information impressed upon an optical pulse is encoded in the internal degrees of freedom of a dense atomic ensemble [39, 40]. Storage times up to 1 s for 20- μ s-long pulses have been measured using this approach [41]. Although exciting, this technique has one main drawback: The frequency of the optical pulses must match the resonance frequency of the atomic ensemble. The use of narrow resonances produced by a dynamically controlled microring resonator is another approach that has been theoretically proposed as a possible solution of this limitation [42].

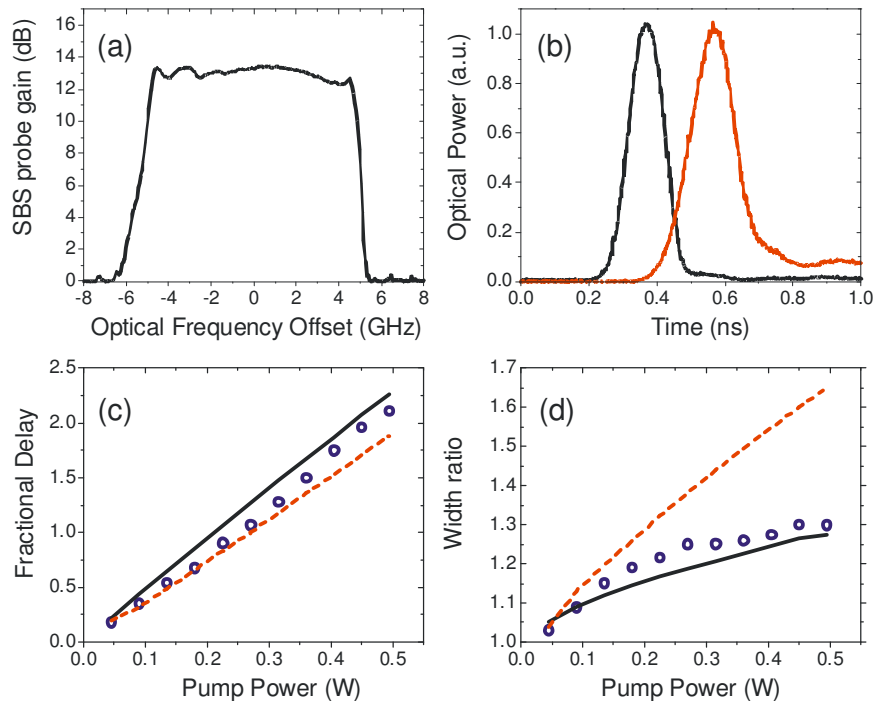


Fig. 4. (a) Optimized gain profile of 9.2 GHz of bandwidth (FWHM) (b) Reference pulse (black line) and delayed pulse (red line) for a pump power of 495 mW. The fractional delay is 2.1 and the output-to-input width ratio is 1.3. (c) Fractional delay vs. input pump power for 100-ps-long pulses. (d) Output-to-input width ratio vs input pump power. Simulations for an optimum flat-top gain profile (black line) and for a Gaussian profile (red dotted line) are also shown. Reprinted figure with permission from Ref. [37].

Stimulated Brillouin scattering has also proven to be able to coherently transform the information contained in the optical pulses to an acoustic excitation for a tunable storage time [43]. As it is the case of SBS slow light, this technique can be implemented at room temperature and works at any wavelength where the fiber is transparent. Figure 5 shows the process of data storage. Data pulses and a short intense “write” pulse initially counter-propagate through the optical fiber, with the frequency of the write pulse downshifted from the data-pulse frequency by the Brillouin frequency of the material (see Fig. 5A). As explained in Section II, in this configuration, the data pulses lie in the anti-Stokes absorption band created by the write pulse. Then, depletion of the data pulses takes place, and an acoustic excitation is created, which retains the information of the data pulses, as it is shown in Fig. 5B. In the retrieval process, a “read” pulse with the same frequency of the write pulse depletes the acoustic wave and converts the data back to the original optical frequency (see Fig. 5C and D).

To optimize the efficiency of the storage and retrieved process, it is useful to consider the “pulse

area” of the write and read pulses, which is defined as $\theta = \kappa \int A(t) dt$, where $A(t)$ is the slowly-varying envelope of the electric field and κ is a proportionality constant depending on the SBS properties of the material. The amplitude of the retrieved data pulses are proportional to the product $\sin(\theta_w)\sin(\theta_r)$, where $\theta_{w(r)}$ refers to the write (read) pulses. Therefore, maximum efficiency is obtained when $\theta_r = \theta_w = (m+1/2)\pi/2$. Besides this condition, in order to obtain a 100% of efficiency, three more requirements must be met. First, the storage time must be less than the acoustic lifetime, which is 3.4 ns for silica. The efficiency of data retrieval decreases exponentially when this condition is not satisfied. Second, the temporal duration of the write and read pulses must be shorter than the shortest duration of the data pulses. In this way, the entire spectrum of the data pulses interact with the SBS resonances created by the write and read pulses, and thus can be stored and retrieved efficiently. Third, the spatial extent of the data packet must be less than twice the length of the optical fiber. This imposes a limit in the number of bits that can be stored.

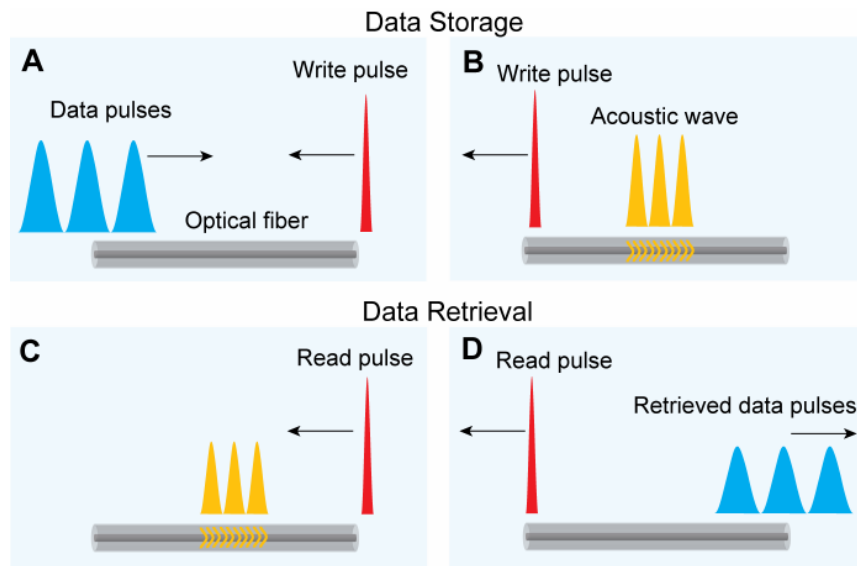


Fig. 5. Storage and retrieval of data pulses in the optical fiber. (A) A short intense write pulse with the frequency downshifted from the data-pulse frequency by the Brillouin frequency of the fiber causes the data pulses to be depleted. An acoustic excitation is then created (B) with the information of the data pulses. (C) A short intense read pulse with the same frequency of the write pulse depletes the acoustic wave and retrieval of the data pulses at the original optical frequency is obtained. From Ref. [43], reprinted with permission from AAAS.

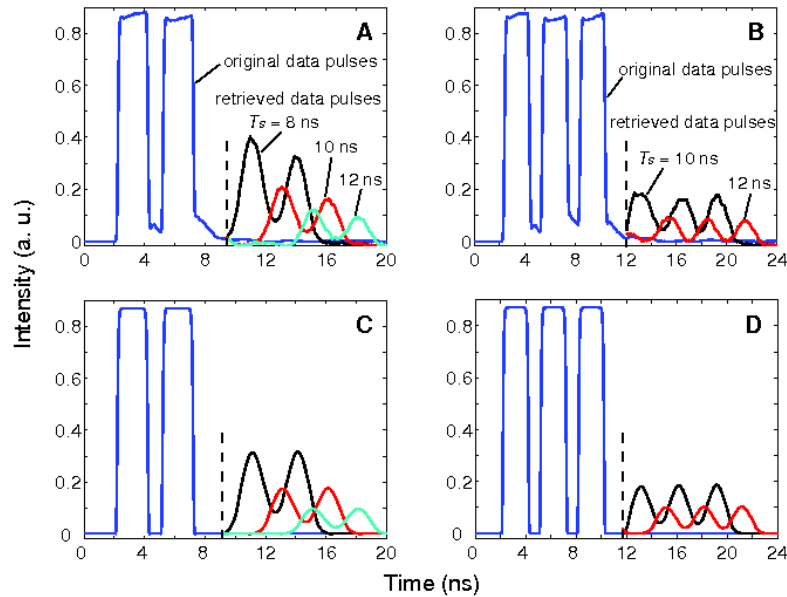


Fig. 6. Retrieved pulse sequences of two (A) and three (B) 2-ns-long pulses separated by 1 ns for different storage times. (C) and (D) show the corresponding theoretical simulations. The retrieved pulses are shown with a magnification factor of 5 to the right of the vertical dashed line. From Ref. [43], reprinted with permission from AAAS. Storage times of 20 times the pulse length with peak powers of the write and read pulses of less than 1 W are feasible using present-day available materials.

Figure 6 shows the measured retrieved pulses for a series of two and three 2-ns-long pulses for different storage times. It can be seen that the efficiency of the process diminishes as the storage time is increased. Readout efficiencies of 29% were achieved for a storage time of 4 ns, while it decreased to 2% for a storage time of 12 ns. The corresponding theoretical simulations are also shown in Figs. 6(C) and (D). As discussed above, SBS stored light in standard optical fibers has great advantages, but it has also some limitations. The limited storage time makes this technique useful for high-speed all-optical processing applications, such as pulse correlation [44] rather than all optical memories. Besides that, the power of the read and write pulses needed to completely deplete the data pulses are of the order of ~ 100 W, which is not low enough for practical applications. Using other materials with longer acoustic lifetimes would extend the storage time while a greater Brillouin gain coefficients or a smaller optical fiber core size would lower the optical power required. Some candidates are chalcogenide fibers (12 ns acoustic lifetime and 134 times the gain coefficient of a standard optical fiber), which have already shown to improve the SBS slow light performance [45], or high pressured gases (80 ns acoustic lifetime for Xe at 140 atm.). Such improvements could make this method suitable for buffering and all-optical processing of information.

6. Future perspectives

Slow light via SBS has shown great flexibility and capacity to reach the high data bit rates needed for use in present telecommunications systems. The possibility of broadening and tailoring the Brillouin gain resonance makes it possible to reduce the distortion while obtaining up to several pulses of delay. However, implementation of an optical buffer based on SBS slow light, is unrealistic at the present time. On the other hand, coherently storing of light based on SBS shows great possibilities that can be exploited in a near future. Tunable storage times of 20 times the pulse length with peak powers of the write and read pulses of less than 1 W are feasible using present-day available materials.

There are other applications to communications systems that do not require more than 1 bit of delay and where SBS slow light has shown to provide flexible solutions. Zhang *et al* [46] recently demonstrated independent delay control and bit-level synchronization of multiple channels using a single slow-light delay line based on SBS. The key factor in this experiment is the ability to generate multiple broadened SBS resonances in an optical fiber using two or more pump beams with different wavelengths. Therefore, each data channel can be independently delayed and controlled by changing the power of the corresponding pump beam. Other examples that benefit from 1-bit delays are optical

correlation and fast optical switching [48], which usually relies in a change of the optical phase, which can be enhanced by the slow-light medium. Besides that, spectral hole-burning of the broadened SBS resonance, recently reported in Ref. [49], opens new possibilities for extremely high density, highly multiplexed data storage.

Applications of slow-light systems are not limited to communications networks. Slowing down the velocity of pulses in a medium can also enhance nonlinear effects, which otherwise would require high optical powers. Enhancements that scale as the square of the slowing factor [5,6] has already been demonstrated. Spectroscopic interferometric measure-

ments can also increase their resolution when a slow-light material is inserted in the interferometer [3,4] with improvement factors up to ~ 100 in Fourier transform interferometers recently reported in Ref. [47]. All of these applications makes SBS slow light a very exciting field of research.

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